

Joint Mitigation of Algorithmic and Physical Errors in Noisy Hamiltonian Simulation

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(Dated:)

Product-formula Hamiltonian simulation is naturally suited to near-term quantum processors, but its accuracy is set by two competing errors: finite-step Trotter bias and physical hardware noise. We introduce a joint extrapolation strategy that ties the tunable noise strength to the Trotter step size, $\lambda(s) = c(sT)^{p+1}$ for a p -th order product formula. Along this one-dimensional path, the leading physical-noise and Trotter terms appear at the same order and can be canceled by a single Richardson extrapolation. The analysis uses a finite-order Baker–Campbell–Hausdorff truncation bound, controlled by doubly right-nested commutators, so that the guarantee does not rely on convergence of the full BCH series. For local Hamiltonians with local Lindbladian noise, the protocol mitigates physical noise together with Trotter error with only a constant asymptotic overhead relative to noiseless Trotter extrapolation, provided the required noise strengths lie above the intrinsic device-noise floor. We demonstrate the workflow on Trotterized Ising dynamics using an adaptively learned sparse Pauli–Lindblad noise strength on a superconducting processor, and benchmark the same one-dimensional construction against a two-dimensional Richardson baseline in tensor-network simulations.

Shuo: To be a physics facing paper

Hamiltonian simulation is a leading route to useful quantum computation, with applications from condensed-matter dynamics to quantum chemistry and high-energy physics [1–4]. Product formulas [5–7] are especially attractive on near-term processors because they compile into repeated local gate layers and require no large ancillary registers. They also retain the locality and commutator structure of the target Hamiltonian, a feature that is often lost in more elaborate primitives such as linear combination of unitaries (LCU) [8, 9] and quantum signal processing (QSP) methods [10, 11].

The practical obstacle is that the two main errors respond oppositely to circuit depth. Increasing the number of Trotter steps reduces the deterministic product-formula bias, but the deeper circuit accumulates more hardware noise [12]. Using fewer steps protects the device from noise, but leaves a larger algorithmic error. In the experimental regime of interest, both effects are visible. It is therefore unnatural to regard one of them as a small correction after the other has already been removed [13].

Standard mitigation treats the two directions separately. One may perform zero-noise extrapolation (ZNE) [14, 15] at fixed Trotter step number and then extrapolate in the step size [16, 17], or sample a two-dimensional Richardson grid over noise strength and step size [18]. The latter approach is systematic, but the number of circuit families grows as a tensor product of the two extrapolation axes. Moreover, large Richardson coefficients can make every grid point expensive to estimate. Here we instead use the depth/noise trade-off to define the extrapolation path itself. We couple the noise strength to the step size so that the leading physical-noise and Trotter terms enter the same expansion variable, and then apply a single Richardson rule along that path. The theory analyzes this coupled expansion using finite BCH truncation, while the experiments below

test how the construction composes with learned-channel ZNE on Trotterized Ising dynamics.

I. RESULTS

Joint Extrapolation Path

Let $H = \sum_{\gamma=1}^{\Gamma} H_{\gamma}$ and suppose that a p -th order product formula implements one short-time step through

$$P(t) = \prod_{v=1}^M e^{-ita_v H_{\gamma_v}}. \quad (1)$$

Here H_{γ} are the Hamiltonian summands used by the product formula, γ_v specifies which summand appears in the v -th exponential and a_v is the corresponding product-formula coefficient, including the fractional weights and repetitions that occur in higher-order formulas. **Xiaoyang: What is H_{γ_v} and a_v ? In the presence of layerwise Markovian noise** We assume that the noise after each Trotter layer can be described by a Markovian Lindbladian process, the corresponding noisy channel is

$$\tilde{\mathcal{P}}(t, \lambda) = \prod_{v=1}^M (e^{\lambda \mathcal{L}_v} e^{-ita_v \text{ad}_{H_{\gamma_v}}}), \quad (2)$$

where λ is a noise strength **Xiaoyang: that can be continuously tuned using, e.g., probabilistic error amplification [1]**, and $\mathcal{L}_v$ is a Lindblad operator. Operationally, λ is the calibrated noise strength supplied by the mitigation procedure: in the hardware workflow below it is implemented by scaling an adaptively learned sparse Pauli–Lindblad channel rather than by increasing the Trotter circuit depth. For step size s and total time T , the measured expectation is

$$g(s, \lambda) = \text{Tr} \left[\mathcal{O} \tilde{\mathcal{P}}(sT, \lambda)^{1/s} \rho_0 \right]. \quad (3)$$

The target value is $\lim_{s, \lambda \rightarrow 0} g(s, \lambda)$.

We set

$$\lambda(s) = c(sT)^{p+1}. \quad (4)$$

Xiaoyang: We can draw a 2-d lattice and this draw this curve for $p = 1$. This choice matches the leading physical-noise contribution, which is linear in λ , to the leading Trotter contribution, which scales as s^{p+1} . Richardson extrapolation applied to the one-variable function $f(s) = g(s, \lambda(s))$ can therefore cancel the leading terms from both sources at once. **Xiaoyang:** Explain Richardson extrapolation in more detail. What is Richardson node and Richardson order? They are used in the remaining text. Here the Richardson order m is the number of extrapolation nodes and fixes how many low-order powers are canceled; in the coupled construction, each node is an experimentally realized pair $(s_j, \lambda(s_j))$ on the curve in Eq. (4) rather than a point in a rectangular two-dimensional grid. See Supplementary Note 1 [19] for the node construction and coefficient bounds. The coupling constant c sets the relative scale of the amplified noise and can be chosen to balance the physical-noise and Trotter remainders. Choosing a smaller exponent would leave noise as the dominant error; choosing a larger exponent would make the procedure behave like ordinary Trotter extrapolation with insufficient noise cancellation. The matched exponent is the point at which the two leading biases enter the same polynomial expansion.

The reduction from a grid to a path also changes the sampling problem. A two-dimensional Richardson rule with m step-size nodes and m' noise nodes requires mm' circuit families before shot allocation is considered. Moreover, if the noise-direction coefficients have one-norm $\|\mathbf{b}'\|_1$, then each grid point must be estimated to accuracy smaller by this factor. For non-optimized grids this norm can grow rapidly with the extrapolation order, creating an additional sampling overhead. Along the coupled path, the coefficients are attached to a single list of experimentally realized pairs $(s_j, \lambda(s_j))$. The protocol is feasible when these nodes remain above the intrinsic device-noise floor. **Xiaoyang:** ? while staying in the perturbative regime where the learned noise model can implement calibrated noise strengths. The lower side of this window is set by the native hardware noise, while the upper side is set by the validity of the learned local noise model; see Supplementary Notes 1 and 4 [19]. This feasibility condition is also the point at which the theoretical construction meets the experiment. The smallest extrapolation node asks for the weakest effective noise, while the largest node asks for the coarsest Trotter step. If the device noise cannot be reduced below the required value, the path has to be shifted upward and the remaining intrinsic noise appears as an uncanceled bias. If the largest amplified-noise point leaves the perturbative regime, the learned channel no longer supplies a reliable local noise model. Thus the method is not a claim that noise can be tuned arbitrarily; it is a way to use the tunable

window that is actually available as efficiently as possible.

Error Cancellation from BCH Truncation

The analysis of the coupled path reduces to one question: after setting $\lambda(s) = c(sT)^{p+1}$, does the observable expectation

$$f(s) = g(s, \lambda(s)) \quad (5)$$

admit a finite expansion whose low-order terms can be canceled by Richardson extrapolation? For noisy product formulas this expansion is obtained from the effective generator of one noisy Trotter step. Instead of assuming convergence of the full BCH series, we use the finite-order BCH truncation bound of Ref. [20] as a tool. For a chosen truncation order q_0 , the noisy one-step channel is replaced by the exponential of its BCH series through order q_0 , and the difference from the true product channel is bounded directly.

This finite truncation is enough for extrapolation. After substituting $\lambda(s) = c(sT)^{p+1}$ and rescaling from one step to total time T , the truncated effective generator has the form

$$\mathcal{G}_{(q_0)}(s) = -i\text{ad}_H + \sum_{q=p}^{m-1} \mathcal{E}_{q+1}(sT)^q + \mathcal{R}_{m,q_0}(s). \quad (6)$$

The first correction terms contain both product-formula error and physical-noise error. The coupling in Eq. (4) is chosen so that the leading noise contribution and the leading Trotter contribution appear at the same order in s . An m -node Richardson rule then cancels the polynomial part through order $m - 1$ with a single set of coefficients, rather than using independent coefficients in a two-dimensional grid.

The final error has two sources. The first is the Richardson remainder $\mathcal{R}_{m,q_0}(s)$, coming from BCH monomials whose effective order is not canceled by the chosen extrapolation nodes. The second is the BCH truncation error from replacing the true noisy product channel by the finite-order effective generator. The truncation theorem bounds this second contribution without requiring the infinite BCH series to converge. For k -local Hamiltonians with local Lindbladian noise, the doubly right-nested commutator bounds from the truncation theorem imply that one can choose $q_0 = O(\log(N/\varepsilon))$ and Richardson order $m = O(\log(N/\varepsilon))$ to obtain an ε -accurate estimator with the same asymptotic scaling as noiseless Trotter extrapolation, up to a constant factor in the step size, provided the required noise strengths remain above the intrinsic noise floor. The supplementary material gives the truncation bookkeeping and model-dependent commutator estimates; the main point here is that the BCH truncation bound converts the coupled noisy evolution into a controlled finite expansion on which Richardson cancellation acts simultaneously on Trotter and noise terms.

Joint Mitigation in Practice

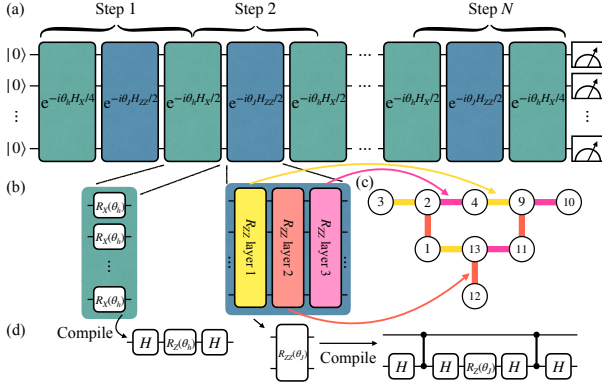


FIG. 1. Hardware implementation of Trotterized Ising dynamics. (a) A second-order symmetric Trotter step alternates transverse-field rotations and Ising interactions. (b) The interaction layer is decomposed into three parallel R_{zz} patterns. (c) A connected 9-qubit subgraph of the 13-qubit device is used; colored edges show the scheduled interaction patterns. (d) Logical R_x and R_{zz} operations are compiled into the native H , R_z , and controlled- Z gates.

We demonstrate the mitigation workflow on transverse-field Ising dynamics,

$$H = -J \sum_{\langle i,j \rangle} Z_i Z_j - h \sum_{i=1}^n X_i, \quad (7)$$

where $\langle i,j \rangle$ denotes the Ising interaction edges in the selected hardware subgraph or in the simulated one-dimensional chain. In the hardware implementation, we equivalently parameterize the transverse-field and interaction strengths by rotation rates $r_x = -h$ and $r_{zz} = -J$. We use a second-order symmetric Trotter formula. The measured observable is the average longitudinal magnetization,

$$O = \frac{1}{n} \sum_{i=1}^n Z_i. \quad (8)$$

The hardware experiments use a connected 9-qubit subgraph of a 13-qubit Tianji S2 chip. **Xiaoyang: We can draw the 9 qubits out of the 13 qubits, Xiaoyang: as shown in Figure 1**, while the tensor-network simulations use the same observable on a 20-qubit one-dimensional instance.

For the hardware data, we model the device noise relevant to the compiled circuit family by an effective sparse Pauli-Lindblad model,

$$\mathcal{L}(\rho) = \sum_{k \in \mathcal{K}} \lambda_k (P_k \rho P_k - \rho), \quad (9)$$

with a retained Pauli set $\mathcal{K}$ much smaller than the full Pauli group. The rates are obtained by an adaptive learning procedure and are used only to set calibrated noise strengths for mitigation, not as a microscopic model of every hardware error process. Scaling the learned rates and sampling the corresponding Pauli insertions provides

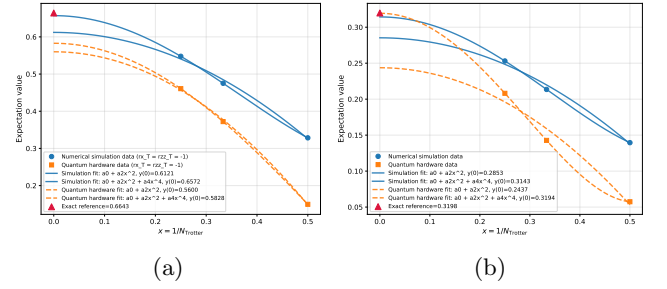


FIG. 2. Joint hardware extrapolation of noise and Trotter errors. Numerical simulation (blue) and superconducting-hardware data (orange) are plotted versus $x = 1/N_{\text{Trotter}}$. The fitted curves extrapolate toward $x = 0$, where both finite-step Trotter error and the amplified physical-noise bias are removed in the fitted model; the red triangle marks the exact reference. Panels show two representative evolution settings: (a) $r_x T = r_{zz} T = -1$ and (b) $r_x T = r_{zz} T = -2$. The compatible hardware and numerical trends support the composability of learned-channel ZNE with Trotter-step extrapolation on the tested device.

the noise strengths for ZNE without increasing the depth of the target Trotter circuit. At fixed Trotter configuration, ZNE uses these calibrated noise strengths to infer the zero-noise response, and the joint protocol then uses the selected configurations as nodes of the one-dimensional extrapolation path. Details of the circuit mapping, readout calibration and learned sparse Pauli-Lindblad model are given in Supplementary Notes 3 and 4 [19].

a. Hardware extrapolation. The learned-noise workflow first enables ZNE at fixed Trotter step number. We then combine the zero-noise estimates across different step numbers and fit in $x = 1/N_{\text{Trotter}}$, exploiting the even-power structure of the second-order product formula. The key experimental check is whether hardware and numerical data move consistently toward the same zero-noise, zero-step reference. This test is not meant to verify the asymptotic scaling theorem by itself; rather, it shows that learned-channel ZNE and Trotter-step extrapolation can be composed on real hardware in the regime where both errors are visible.

Xiaoyang: Figure 2 shows the expectation values of the average magnetization versus the inverse Trotter steps, where each plotted configuration corresponds to a data point on the one-dimensional joint extrapolation path and the final estimate is obtained by the resulting 1D extrapolation. The subsequent fit in $x = 1/N_{\text{Trotter}}$ implements the one-dimensional joint error-mitigation step: the learned-channel ZNE removes the physical-noise bias at fixed Trotter step number, and the remaining inverse-step extrapolation removes the finite-step Trotter bias along the same one-dimensional mitigation workflow. The two parameter settings change both the target evolution angle and the depth-dependent balance between Trotter and noise effects, so agreement across the two panels supports the stability of the 1D

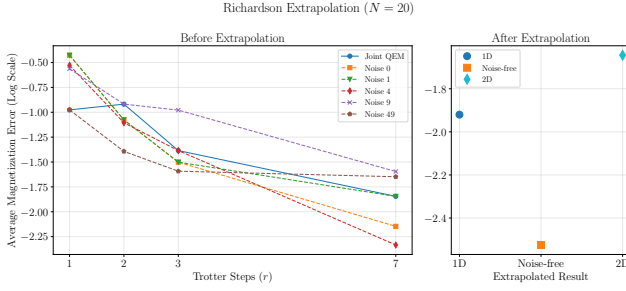


FIG. 3. Tensor-network validation on a 20-qubit one-dimensional Ising instance. Average magnetization error versus the number of Trotter steps. The one-dimensional joint method reduces the bias of noisy estimates and gives a final extrapolated value close to the two-dimensional Richardson baseline.

joint error-mitigation workflow on the tested device.

b. Tensor-network validation. We also study the joint extrapolation path directly in a controlled tensor-network simulation using `quimb` [21]. The one-dimensional protocol evaluates the coupled curve $\lambda(s) = c(sT)^{p+1}$, while the baseline uses a tensor-product Richardson rule over independent step-size and noise-strength nodes. Thus the two-dimensional baseline provides a useful accuracy benchmark, but requires a rectangular grid of noisy circuit settings. **Xiaoyang:** As shown in Figure 3, the joint one-dimensional extrapolation has an evaluation error of $10^{-1.9}$, which is smaller than the two-dimensional baseline while using a one-dimensional set of extrapolation nodes while the measurement cost is reduced to one-fifth of the original. **Xiaoyang:** Therefore, the one-dimensional protocol preserves the extrapolation accuracy while reducing the measurement cost.

II. DISCUSSION

The commutator analysis, hardware implementation, and numerical comparison indicate that physical noise can be incorporated into the same extrapolation variable that controls Trotter error when reliable tunable noise strengths are available. The remaining questions are practical: how to optimize the coupling constant and nodes for a given device, how accurately the effective Lindbladian can be learned at larger depths, and how the method behaves under non-Markovian or strongly correlated noise. These questions set the natural next scale-up path. The method is suitable for near-term and early fault-tolerant devices where increasing circuit depth is algorithmically urgent but physically unfeasible.

III. METHODS

a. Richardson nodes. For a function

$$f(s) = f_0 + \sum_{\ell=1}^{m-1} a_{\ell} s^{\ell} + R_m(s), \quad (10)$$

an m -node Richardson rule estimates f_0 by $\sum_{j=1}^m b_j f(s_j)$, where the coefficients satisfy $\sum_j b_j = 1$ and $\sum_j b_j s_j^{\ell} = 0$ for $1 \leq \ell \leq m-1$. In the coupled protocol, the measured nodes are the pairs $(s_j, \lambda(s_j))$ with $\lambda(s) = c(sT)^{p+1}$. We choose $s_j = T/r_j$ with integer Trotter step counts r_j , so every node corresponds to an executable product-formula circuit. The residual bias is bounded by $\|\mathbf{b}\|_1 \max_j |R_m(s_j)|$.

The resource estimates use the logarithmic-norm Richardson construction from product-formula extrapolation, which keeps $\|\mathbf{b}\|_1$ logarithmic in the extrapolation order while keeping the largest step size within the finite-BCH control regime. The hardware analysis uses the same one-dimensional logic in the form available on the device: learned-channel ZNE is first applied at fixed N_{Trotter} , and the resulting estimates are then extrapolated in $x = 1/N_{\text{Trotter}}$. For the second-order symmetric formula used in the experiment, the fit uses even powers of x .

Sampling costs follow directly from the Richardson coefficients. If $\hat{f}_0 = \sum_j b_j \hat{f}(s_j)$ and the node estimates are independent, then the variance is $\sum_j b_j^2 \sigma_j^2$. Thus, for comparable per-shot variances, the shot cost scales with $\|\mathbf{b}\|_2^2$ up to constants. A two-dimensional Richardson baseline replaces b_j by tensor-product coefficients $b_j b'_{\ell}$ and requires a rectangular set of circuit families. The coupled path removes this second extrapolation axis.

b. Finite-order BCH control. For one step of the noisy product formula we use generators

$$\mathcal{K}_{2v-1} = -isT a_v \text{ad}_{H_{q_v}}, \quad \mathcal{K}_{2v} = c(sT)^{p+1} \mathcal{L}_v. \quad (11)$$

The proof does not assume that the full BCH series converges. Instead, it expands the one-step channel to a finite order q_0 and bounds the discarded tail directly using the truncation theorem of Ref. [20]. The tail is controlled by doubly right-nested commutator sums of the schematic form

$$\sum \left\| \left[[\mathcal{K}_{v_{1,1}}, \dots, \mathcal{K}_{v_{1,q_1}}], \dots, [\mathcal{K}_{v_{d,1}}, \dots, \mathcal{K}_{v_{d,q_d}}] \right] \right\|. \quad (12)$$

After substituting $\lambda(s) = c(sT)^{p+1}$, each BCH monomial is graded by its net power of s in the continuous-time effective generator. A Hamiltonian factor contributes one power of sT , a noise factor contributes $(p+1)$ powers through $\lambda(s)$, and passing from one step to total time divides by sT . Consequently, the ideal Hamiltonian term has grade zero, while the leading product-formula and physical-noise corrections both have grade p . Richardson cancellation through grade $m-1$ therefore removes the two leading mechanisms with the same coefficients.

The remaining bias has two pieces: BCH monomials whose grade is not canceled by the chosen Richardson rule, and the finite-order truncation error from discarding terms above q_0 . For k -local Hamiltonians and local Lindbladian noise, nested commutators are nonzero only when successive local supports overlap. The commutator sums are therefore governed by support growth and local-extensiveness parameters, rather than by the extensive operator norm of the full Hamiltonian. Taking $q_0 = O(\log(N/\epsilon))$ makes the truncation error $O(\epsilon)$ while changing the allowed step size only by logarithmic local factors. For plane-wave electronic-structure Hamiltonians, the corresponding control parameter is the layer number of the second-quantized monomials generated under repeated commutation. The supplementary material gives the full truncation bookkeeping; here we use it only to justify the finite expansion on which the coupled Richardson rule acts.

c. Intrinsic noise floor. The theoretical path assumes that the experiment can realize the effective noise strengths $\lambda(s_j)$ at all Richardson nodes. Real hardware has an intrinsic noise level λ_0 before amplification, so the implemented values are better viewed as $\lambda_0 + \lambda_{\text{amp}}(s_j)$. If λ_0 is comparable to the smallest target value on the path, the uncanceled offset contributes a residual term that is not removed by the coupled Richardson coefficients. The resource statement therefore assumes that the floor is below the smallest coupled noise strength up to the target additive accuracy. In practice, one first chooses the noise-strength window accessible to the device and then selects c and the Richardson nodes so that every $\lambda(s_j)$ lies inside that window. The same tunable noise strength is also required by a two-dimensional baseline; the coupled construction changes how many noise-strength values must be sampled.

d. Learned noise model. The hardware workflow uses an adaptively learned effective sparse Pauli–

Lindblad model

$$\mathcal{L}(\rho) = \sum_{k \in \mathcal{K}} \lambda_k (P_k \rho P_k - \rho), \quad (13)$$

with $|\mathcal{K}|$ much smaller than the full Pauli group. The Pauli–Lindblad rates are obtained by adaptive learning on circuits matched to the target compiled circuit family. The corresponding channel is sampled through random Pauli insertions, which implements calibrated noise strengths for ZNE without lengthening the target Trotter circuit. The learned channel is used as an effective model for the observable and circuit family; it is not intended to reconstruct every microscopic noise process.

e. Hardware data processing. The measured observable is the average longitudinal magnetization, estimated from bitstrings produced by computational-basis measurements. Readout mitigation is applied before gate-noise learning by inverting local calibration matrices for the selected qubits at the observable level. For each Trotter step number and noise-strength setting, repeated circuit executions provide sample means and uncertainty estimates. Zero-noise estimates are obtained by fitting the noise-strength dependence with low-order polynomial or exponential ansatz functions. The corrected values are then fitted as even polynomials in $1/N_{\text{Trotter}}$ for the second-order symmetric formula.

f. Classical simulations. The tensor-network data use the same average magnetization observable as the hardware experiment, but on a 20-qubit one-dimensional transverse-field Ising instance. Noisy and noiseless product-formula circuits are represented with `quimb`. Noise is inserted as a layerwise channel with noise strengths chosen to match the extrapolation nodes. The one-dimensional protocol evaluates the coupled curve $\lambda(s) = c(sT)^{p+1}$, whereas the baseline evaluates a rectangular grid in (s, λ) . Both methods are compared against the same noiseless reference expectation, so the plotted errors isolate the bias removed by extrapolation rather than differences in state preparation or measurement.

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- [1] Youngseok Kim, Andrew Eddins, Sajant Anand, Ken Xuan Wei, Ewout Van Den Berg, Sami Rosenblatt, Hasan Nayfeh, Yantao Wu, Michael Zaletel, Kristan Temme, et al., *Evidence for the utility of quantum computing before fault tolerance*, *Nature* **618**, no. 7965, 500–505 (2023).
 - [2] Andrew M. Childs, Dmitri Maslov, Yunseong Nam, Neil J. Ross, and Yuan Su, *Toward the first quantum simulation with quantum speedup*, *Proceedings of the National Academy of Sciences* **115**, no. 38, 9456–9461 (2018).
 - [3] Sam McArdle, Suguru Endo, Alán Aspuru-Guzik, Simon C. Benjamin, and Xiao Yuan, *Quantum computational chemistry*, *Reviews of Modern Physics* **92**, no. 1, 015003 (2020).
 - [4] Christian W. Bauer, Zohreh Davoudi, A. B. Balantekin, Tanmoy Bhattacharya, Marcela Carena, Wibe A. de Jong, Patrick Draper, Aida X. El-Khadra, Nathan Gemelke, Masanori Hanada, Dmitri Kharzeev, Henry Lamm, Ying-Ying Li, Junyu Liu, Mikhail Lukin, Yannick Meurice, Christopher Monroe, Benjamin Nachman, Guido Pagano, John Preskill, Enrico Rinaldi, Alessandro Roggero, David I. Santiago, Martin J. Savage, Irfan Siddiqi, George Siopsis, Drew Van Zanten, Nathan Wiebe, Yukari Yamauchi, Kubra Yeter-Aydeniz, and Silvia Zorzetti, *Quantum simulation for high-energy physics*, *PRX Quantum* **4**, no. 2, 027001 (2023).
 - [5] Seth Lloyd, *Universal quantum simulators*, *Science* **273**, no. 5278, 1073–1078 (1996).
 - [6] Masuo Suzuki, *Decomposition formulas of exponential operators and Lie exponentials with some applications*

- to quantum mechanics and statistical physics, *Journal of Mathematical Physics* **26**, no. 4, 601–612 (1985).
- [7] Masuo Suzuki, *General theory of fractal path integrals with applications to many-body theories and statistical physics*, *Journal of Mathematical Physics* **32**, no. 2, 400–407 (1991).
 - [8] Andrew M. Childs and Nathan Wiebe, *Hamiltonian simulation using linear combinations of unitary operations*, *Quantum Information and Computation* **12**, no. 11-12, 901–924 (2012).
 - [9] Dominic W. Berry, Andrew M. Childs, Richard Cleve, Robin Kothari, and Rolando D. Somma, *Simulating Hamiltonian dynamics with a truncated Taylor series*, *Physical Review Letters* **114**, no. 9, 090502 (2015).
 - [10] Guang Hao Low and Isaac L. Chuang, *Optimal Hamiltonian simulation by quantum signal processing*, *Physical Review Letters* **118**, no. 1, 010501 (2017).
 - [11] Guang Hao Low and Isaac L. Chuang, *Hamiltonian simulation by qubitization*, *Quantum* **3**, 163 (2019).
 - [12] Jue Xu, Chu Zhao, Junyu Fan, and Qi Zhao, *Exponentially decaying quantum simulation error with noisy devices*, 2025, [2504.10247](#).
 - [13] Shigeo Hakkaku, Yasunari Suzuki, Yuuki Tokunaga, and Suguru Endo, *Data-efficient error mitigation for physical and algorithmic errors in a Hamiltonian simulation*, 2025, [arXiv:2503.05052](#).
 - [14] Ying Li and Simon C Benjamin, *Efficient variational quantum simulator incorporating active error minimization*, *Physical Review X* **7**, no. 2, 021050 (2017).
 - [15] Kristan Temme, Sergey Bravyi, and Jay M. Gambetta, *Error mitigation for short-depth quantum circuits*, *Physical Review Letters* **119**, 180509 (2017).
 - [16] James D Watson and Jacob Watkins, *Exponentially reduced circuit depths using Trotter error mitigation*, *PRX Quantum* **6**, no. 3, 030325 (2025).
 - [17] Kaoru Mizuta, *On the commutator scaling in Hamiltonian simulation with multi-product formulas*, 2025, [arXiv:2507.06557](#).
 - [18] Almudena Carrera Vazquez, Daniel J Egger, David Ochsner, and Stefan Woerner, *Well-conditioned multi-product formulas for hardware-friendly hamiltonian simulation*, *Quantum* **7**, 1067 (2023).
 - [19] See the Supplementary Information.
 - [20] Xinzhao Wang, Shuo Zhou, Xiaoyang Wang, Yi-Cong Zheng, Shengyu Zhang, and Tongyang Li, *Lindbladian simulation with commutator bounds*, 2026, [arXiv:2603.28602](#).
 - [21] Johnnie Gray, *quimb: A python package for quantum information and many-body calculations*, *Journal of Open Source Software* **3**, no. 29, 819 (2018).